Final Exam B Solutions: Optimization, Lagrangians, and Dynamic Equations

GSE 510

Solutions

1. **Unconstrained Optimization: Local Maximum**

Consider the function
$$f(x, y) = 24x + 18y - 3x^2 - 2y^2.$$

(a) The first partial derivatives are
$$f_x = 24 - 6x$$
and
$$f_y = 18 - 4y.$$

(b) The critical point solves
$$24 - 6x = 0$$
and
$$18 - 4y = 0.$$
Therefore,
$$x = 4$$
and
$$y = \frac{18}{4} = \frac{9}{2}.$$
The critical point is
$$\left(4, \frac{9}{2}\right).$$

(c) The second partial derivatives are
$$f_{xx} = -6, \qquad f_{yy} = -4, \qquad f_{xy} = 0.$$
Therefore, the Hessian matrix is
$$H = \begin{bmatrix} -6 & 0 \\ 0 & -4 \end{bmatrix}.$$

(d) The determinant of the Hessian is
$$|H| = (-6)(-4) - 0(0) = 24.$$
Since
$$|H| > 0$$
and
$$f_{xx} = -6 < 0,$$
the critical point is a local maximum.

(e) The value of the function at the critical point is

$$f\left(4, \frac{9}{2}\right) = 24(4) + 18\left(\frac{9}{2}\right) - 3(4)^2 - 2\left(\frac{9}{2}\right)^2.$$

Compute:

$$24(4) = 96,$$

$$18\left(\frac{9}{2}\right) = 81,$$

$$3(4)^2 = 48,$$

and

$$2\left(\frac{9}{2}\right)^2 = 2\left(\frac{81}{4}\right) = \frac{81}{2}.$$

Therefore,

$$f\left(4, \frac{9}{2}\right) = 96 + 81 - 48 - \frac{81}{2} = 129 - \frac{81}{2} = \frac{177}{2}.$$

2. **Unconstrained Optimization: Saddle Point**

Consider

$$f(x, y) = x^2 - y^2 + 4x + 6y.$$

(a) The first partial derivatives are

$$f_x = 2x + 4$$

and

$$f_y = -2y + 6.$$

(b) The critical point solves

$$2x + 4 = 0$$

and

$$-2y + 6 = 0.$$

Therefore,

$$x = -2$$

and

$$y = 3.$$

The critical point is

$$(-2, 3).$$

(c) The second partial derivatives are

$$f_{xx} = 2, \qquad f_{yy} = -2, \qquad f_{xy} = 0.$$

Therefore,

$$H = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}.$$

(d) The determinant of the Hessian is

$$|H| = 2(-2) - 0(0) = -4.$$

(e) Since

$$|H| < 0,$$

the critical point is a saddle point. The function curves upward in the x-direction and downward in the y-direction, so the point is neither a local maximum nor a local minimum.

3. **Profit Maximization with Two Choice Variables**

A firm chooses labor L and capital K. Profit is

$$\pi(L, K) = 80L + 100K - 2L^2 - 4K^2 - 2LK.$$

(a) The first partial derivatives are

$$\pi_L = 80 - 4L - 2K$$

and

$$\pi_K = 100 - 8K - 2L.$$

(b) The first-order conditions are

$$80 - 4L - 2K = 0$$

and

$$100 - 8K - 2L = 0.$$

Rearranging,

$$4L + 2K = 80$$

and

$$2L + 8K = 100.$$

Divide the first equation by 2:

$$2L + K = 40.$$

Divide the second equation by 2:

$$L + 4K = 50.$$

From the first equation,

$$K = 40 - 2L.$$

Substitute into the second equation:

$$L + 4(40 - 2L) = 50.$$

Therefore,

$$L + 160 - 8L = 50,$$

so

$$-7L = -110.$$

Hence,

$$L^* = \frac{110}{7}.$$

Then

$$K^* = 40 - 2\left(\frac{110}{7}\right) = \frac{280}{7} - \frac{220}{7} = \frac{60}{7}.$$

The critical point is

$$\left(\frac{110}{7}, \frac{60}{7}\right).$$

(c) The second partial derivatives are

$$\pi_{LL} = -4, \qquad \pi_{KK} = -8, \qquad \pi_{LK} = -2.$$

Therefore, the Hessian matrix is

$$H = \begin{bmatrix} -4 & -2 \\ -2 & -8 \end{bmatrix}.$$

(d) The determinant is

$$|H| = (-4)(-8) - (-2)(-2) = 32 - 4 = 28.$$

Since

$$|H| > 0$$

and

$$\pi_{LL} = -4 < 0,$$

the critical point is a local maximum.

(e) The cross-partial derivative is

$$\pi_{LK} = -2.$$

This means that increasing capital lowers the marginal profit from labor, and increasing labor lowers the marginal profit from capital. Economically, the two inputs interact negatively in the profit function.

4. **Lagrangian: Consumer Choice**

Consider the problem

$$\max_{x,y} x^{1/3}y^{2/3}$$

subject to

$$x + 2y = 30.$$

(a) The Lagrangian is

$$\mathcal{L}(x, y, \lambda) = x^{1/3}y^{2/3} + \lambda(30 - x - 2y).$$

(b) The first-order conditions are

$$\mathcal{L}_x = \frac{1}{3}x^{-2/3}y^{2/3} - \lambda = 0,$$

$$\mathcal{L}_y = \frac{2}{3}x^{1/3}y^{-1/3} - 2\lambda = 0,$$

and

$$\mathcal{L}_\lambda = 30 - x - 2y = 0.$$

(c) From the first-order conditions,

$$MU_x = \lambda$$

and

$$MU_y = 2\lambda.$$

Therefore,

$$\frac{MU_x}{MU_y} = \frac{1}{2}.$$

Now

$$\frac{MU_x}{MU_y} = \frac{\frac{1}{3}x^{-2/3}y^{2/3}}{\frac{2}{3}x^{1/3}y^{-1/3}} = \frac{1}{2}\frac{y}{x}.$$

Therefore,

$$\frac{1}{2}\frac{y}{x} = \frac{1}{2}.$$

Hence,

$$y = x.$$

(d) Use the budget constraint:

$$x + 2y = 30.$$

Since $y = x$,

$$x + 2x = 30.$$

Therefore,

$$3x = 30,$$

so

$$x^* = 10.$$

Hence,

$$y^* = 10.$$

The optimal bundle is

$$(x^*, y^*) = (10, 10).$$

(e) The Lagrange multiplier measures the marginal value of relaxing the constraint. In this problem, it tells us approximately how much the maximum utility would increase if the right-hand side of the constraint increased by one unit.

5. **Bordered Hessian**

Consider

$$\max_{x,y} xy$$

subject to

$$x + y = 12.$$

(a) The Lagrangian is

$$\mathcal{L}(x, y, \lambda) = xy + \lambda(12 - x - y).$$

(b) The first-order conditions are

$$\mathcal{L}_x = y - \lambda = 0,$$
$$\mathcal{L}_y = x - \lambda = 0,$$

and

$$\mathcal{L}_\lambda = 12 - x - y = 0.$$

From the first two equations,

$$x = y.$$

Using the constraint,

$$x + y = 12,$$

gives

$$2x = 12.$$

Thus,

$$x^* = 6, \qquad y^* = 6.$$

Also,

$$\lambda^* = 6.$$

(c) The constraint is

$$g(x, y) = x + y = 12,$$

so

$$g_x = 1, \qquad g_y = 1.$$

The second derivatives of the Lagrangian with respect to x and y are

$$\mathcal{L}_{xx} = 0, \qquad \mathcal{L}_{yy} = 0, \qquad \mathcal{L}_{xy} = 1.$$

Therefore, the bordered Hessian is

$$B = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

(d) Since the bordered Hessian is constant, evaluating it at the critical point $(6, 6)$ gives

$$B(6, 6) = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

(e) The determinant is

$$|B| = \begin{vmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{vmatrix} = 2.$$

Using the sign convention discussed in class, this bordered Hessian determinant indicates a constrained maximum. This makes sense because along the line $x+y = 12$, the product xy is maximized when x and y are equal.

6. **When the Lagrangian Is Not the Whole Story**

A consumer has utility

$$u(x, y) = x + 3y$$

and budget constraint

$$x + y = 12,$$

with

$$x \geq 0, \qquad y \geq 0.$$

(a) The equality-constrained Lagrangian is

$$\mathcal{L}(x, y, \lambda) = x + 3y + \lambda(12 - x - y).$$

(b) The first-order conditions are

$$\mathcal{L}_x = 1 - \lambda = 0,$$

$$\mathcal{L}_y = 3 - \lambda = 0,$$

and

$$\mathcal{L}_\lambda = 12 - x - y = 0.$$

(c) The first two conditions imply

$$\lambda = 1$$

and

$$\lambda = 3.$$

These cannot both hold. Therefore, the standard interior Lagrangian method does not deliver an interior solution. Economically, good y gives more utility per unit than good x, so the consumer wants to spend everything on good y.

(d) At $(12, 0)$,

$$u(12, 0) = 12 + 3(0) = 12.$$

At $(0, 12)$,

$$u(0, 12) = 0 + 3(12) = 36.$$

(e) The true utility-maximizing bundle is

$$(x^*, y^*) = (0, 12).$$

(f) This is a corner solution because the consumer chooses zero units of good x and spends everything on good y. The optimum occurs at the endpoint of the budget line rather than at an interior tangency point.

7. **Constrained Optimization with Boundary Check**

Consider

$$\max_{x,y} x^2 y$$

subject to

$$x + y = 15,$$

with

$$x \geq 0, \qquad y \geq 0.$$

(a) The Lagrangian is

$$\mathcal{L}(x, y, \lambda) = x^2 y + \lambda(15 - x - y).$$

(b) The first-order conditions for an interior solution are

$$\mathcal{L}_x = 2xy - \lambda = 0,$$

$$\mathcal{L}_y = x^2 - \lambda = 0,$$

and

$$\mathcal{L}_\lambda = 15 - x - y = 0.$$

(c) From the first two first-order conditions,

$$2xy = \lambda$$

and

$$x^2 = \lambda.$$

Therefore,

$$2xy = x^2.$$

For an interior solution, $x > 0$, so divide by x:

$$2y = x.$$

Thus,

$$x = 2y.$$

Use the constraint:

$$x + y = 15.$$

Substituting $x = 2y$,

$$2y + y = 15.$$

Hence,

$$3y = 15,$$

so

$$y^* = 5.$$

Therefore,

$$x^* = 10.$$

The interior critical point is

$$(10, 5).$$

(d) The objective value at the interior critical point is

$$f(10, 5) = 10^2(5) = 100(5) = 500.$$

(e) At the endpoint $(15, 0)$,

$$f(15, 0) = 15^2(0) = 0.$$

At the endpoint $(0, 15)$,

$$f(0, 15) = 0^2(15) = 0.$$

(f) Yes, the interior critical point is the global maximum. The endpoint values are both 0, while the interior critical point gives value 500. Since the feasible set is the closed line segment between the two endpoints, checking the interior critical point and the endpoints is enough for this problem.

8. **Linear Difference Equation**

Consider

$$x_{t+1} = 0.75x_t + 5.$$

(a) A steady state satisfies
$$x^* = 0.75x^* + 5.$$
Therefore,
$$x^* - 0.75x^* = 5,$$
so
$$0.25x^* = 5.$$
Hence,
$$x^* = 20.$$

(b) Suppose $x_0 = 0$. Then
$$x_1 = 0.75(0) + 5 = 5.$$
Next,
$$x_2 = 0.75(5) + 5 = 3.75 + 5 = 8.75.$$
Finally,
$$x_3 = 0.75(8.75) + 5 = 6.5625 + 5 = 11.5625.$$

(c) Since $x^* = 20$, subtract 20 from both sides:
$$x_{t+1} - 20 = 0.75x_t + 5 - 20.$$
Therefore,
$$x_{t+1} - 20 = 0.75x_t - 15.$$
Since
$$0.75(x_t - 20) = 0.75x_t - 15,$$
we can write
$$x_{t+1} - x^* = 0.75(x_t - x^*).$$

(d) The steady state is stable because
$$|0.75| < 1.$$
Deviations from the steady state shrink over time.

(e) If x_t is capital, then the equation says next period's capital depends partly on current capital and partly on a constant addition. The steady state $x^* = 20$ is the long-run capital level. If capital starts below 20, it moves upward toward 20.

9. **Differential Equation and Stability**

Consider
$$\frac{dp}{dt} = 0.4(D(p) - S(p)),$$
where
$$D(p) = 90 - 2p, \qquad S(p) = 30 + p.$$

(a) Substitute demand and supply into the differential equation:
$$\frac{dp}{dt} = 0.4\left[(90 - 2p) - (30 + p)\right].$$

(b) Simplify:
$$(90 - 2p) - (30 + p) = 60 - 3p.$$
Therefore,
$$\frac{dp}{dt} = 0.4(60 - 3p).$$
Hence,
$$\frac{dp}{dt} = 24 - 1.2p.$$

(c) A steady-state price satisfies

$$24 - 1.2p^* = 0.$$

Therefore,

$$1.2p^* = 24,$$

so

$$p^* = 20.$$

(d) The steady state is stable. If

$$p < 20,$$

then

$$24 - 1.2p > 0,$$

so p rises. If

$$p > 20,$$

then

$$24 - 1.2p < 0,$$

so p falls. Thus, price moves toward 20.

(e) The adjustment rule says price rises when demand exceeds supply and falls when supply exceeds demand. The steady state is the market-clearing price where demand equals supply.

10. **Solow-Style Difference Equation**

Consider

$$k_{t+1} = sAk_t^{1/2} + (1-\delta)k_t,$$

where

$$s = 0.5, \qquad A = 8, \qquad \delta = 0.25.$$

(a) Substitute the parameter values:

$$k_{t+1} = 0.5(8)k_t^{1/2} + (1 - 0.25)k_t.$$

Therefore,

$$k_{t+1} = 4k_t^{1/2} + 0.75k_t.$$

(b) A steady state satisfies

$$k^* = 4(k^*)^{1/2} + 0.75k^*.$$

Rearranging,

$$k^* - 0.75k^* = 4(k^*)^{1/2}.$$

Thus,

$$0.25k^* = 4(k^*)^{1/2}.$$

If $k^* > 0$, divide by $(k^*)^{1/2}$:

$$0.25(k^*)^{1/2} = 4.$$

Therefore,

$$(k^*)^{1/2} = 16.$$

Squaring both sides,

$$k^* = 256.$$

There is also a zero steady state:

$$k^* = 0.$$

The positive steady state is

$$k^* = 256.$$

(c) Suppose $k_0 = 64$. Then

$$k_1 = 4\sqrt{64} + 0.75(64).$$

Since

$$\sqrt{64} = 8,$$

we get

$$k_1 = 4(8) + 48 = 32 + 48 = 80.$$

(d) Suppose $k_0 = 400$. Then

$$k_1 = 4\sqrt{400} + 0.75(400).$$

Since

$$\sqrt{400} = 20,$$

we get

$$k_1 = 4(20) + 300 = 80 + 300 = 380.$$

(e) Capital moves toward the positive steady state $k^* = 256$. If

$$k_0 = 64 < 256,$$

then

$$k_1 = 80 > 64,$$

so capital rises toward the steady state. If

$$k_0 = 400 > 256,$$

then

$$k_1 = 380 < 400,$$

so capital falls toward the steady state.

(f) Economically, the positive steady state is the capital level where investment exactly offsets depreciation. At this level, the capital stock remains constant over time.